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Fact-checking misinformation on Chinese Social Media: Impact of corrections, awareness prompts, and legal warnings on endorsement

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Abstract

This study investigates the differential impacts of corrections, awareness prompts, and legal warnings on the endorsement of fact-checking information (through both "likes" and expressed support in associated comments) across three types of misinformation motivation (dread, wedge-driving, wish) on Weibo, a major Chinese social media platform. Through manual labeling and BERT (a pretrained large language model), we analyzed a cleaned dataset of 4,942 original fact-checking Weibo posts from 18 November 2010 to 31 May 2022, created or shared by Weibo Piyao. Results indicate that government posts or those with visual cues received fewer "likes" but garnered more supportive comments, while awareness prompts and legal warnings received more supportive comments across three misinformation types. This research provides valuable insights into the practice of fact-checking on social media, highlighting how different strategies may vary in their impact depending on the nature of the misinformation being addressed.

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Keywords
Awareness prompts, fact-checking, legal warnings, misinformation, social endorsement, social media, Weibo

Social media, while widely criticized for being a conduit for misinformation, also serves as a crucial platform for fact-checking. In the global Chinese community, Weibo is one of most influential platforms in and outside of China. It is a counterpart of Twitter (Now X), serving as a general public sphere where government and media authorities assertively define and combat fake news (Pang et al., 2022). Fact-checking on social media plays an important role in reducing the impact of misinformation on individuals' beliefs (Courchesne et al., 2021) and intentions to share misinformation (Brenes Peralta et al., 2022; Nekmat, 2020). Common strategies for fact-checking misinformation employed worldwide include issuing corrections, using awareness prompts, and providing legal warnings (Poynter, 2023). While the first two are widely used globally, legal warnings are primarily employed in countries like China (Rodrigues and Xu, 2020).

Audience reactions to fact-checking posts on social media, such as "likes" and comments, play a crucial role in evaluating the effectiveness and favorability of these strategies. "Likes" serve as a measure of endorsement level, reflecting interest, approval, and content visibility (Chen et al., 2023; Sumner et al., 2018), while comments provide deeper insights into user opinions, emotions, and suggestions that a simple "likes" cannot capture. These endorsements through "likes" and comments promote the credibility and influence of fact-checking content (Borah and Xiao, 2018; Luo et al., 2022), increasing the likelihood of its selection and dissemination (Messing and Westwood, 2013). Understanding these dynamics is essential for promoting the spread of fact-checking misinformation on social media. However, there is limited research on the potential impacts of corrections, awareness prompts, and legal warnings on endorsement, measured through both "likes" and comments of fact-checking posts.

The effectiveness of fact-checking is not always satisfactory, which highlights the need to examine the underlying motivations driving the spread of misinformation. Misinformation often spreads purposefully by appealing to people's emotional or psychological motivations – such as wish (hope), dread (fear), or wedge-driving (anger) (Chen et al., 2021)—a classification similar to that of rumors (Chua et al., 2017; Knapp, 1944; Song et al., 2021). For instance, wish-driven misinformation leverages optimism to encourage sharing, while dread-driven misinformation exploits anxiety to provoke rapid dissemination. Wedge-driving misinformation specifically targets certain groups or individuals, aiming to create or deepen tensions and conflicts by provoking strong emotional reactions against particular targets, such as police.

Research has shown that emotions like hope, fear, and anger significantly influence how people interact with and share misinformation (Freiling et al., 2021; Lee et al., 2021), demonstrating the insufficiency of a "one size fits all" fact-checking approach. Classifying misinformation based on these motivations may allow for the development of more targeted and effective strategies. These motivations may elicit varied cognitive

and affective responses, as well as subsequent behaviors. However, it remains unexplored whether different strategies have varying impacts on endorsement. Therefore, we aim to investigate how fact-checking strategies influence endorsement across dread, wedge-driving, and wish misinformation. Understanding these dynamics is vital for designing fact-checking strategies tailored to the specific motivations underlying misinformation, ultimately improving their overall effectiveness.

Theoretical framework

Misinformation and fact-checking

Misinformation refers to claims shown to be false or misleading based on the best evidence and expert analysis (Vraga and Bode, 2020). Due to characteristics such as low entry barriers and anonymity, social media platforms often lack stringent content moderation mechanisms. As a result, misinformation can spread widely and rapidly. Fact-checking has emerged as one of the most widely adopted innovations to combat misinformation by systematically evaluating the validity of claims made by public officials and institutions with an explicit attempt to identify whether a claim is factual (Walter et al., 2020).

While fact-checking has become a global response to misinformation, the development of fact-checking practices in China reflects unique challenges and localized adaptations. Compared to Western countries, fact-checking in China started relatively late, with initial efforts around 2010 focusing on debunking misinformation. Platforms such as "Truth Seeking" column by People's Daily Online emerged to provide fact-checking news and conduct post-publication fact-checking of news already released by other media outlets. Although the scale of fact-checking platforms in China is relatively small, it continues to grow. In China, fact-checking posts are more often centered on issues related to everyday life, such as health, food safety, and natural disasters with less emphasis on politics (Liu and Zhou, 2022).

Ensuring fact-checking effectiveness is a complex and multifaceted challenge. Cultural diversity and geopolitical dynamics significantly influence the success of fact-checking strategies. For example, corrections and using awareness prompts are commonly used worldwide (Poynter, 2023), whereas legal warnings are more prevalent in countries like China (Rodrigues and Xu, 2020). In addition, misinformation spans diverse topics, including politics, health, and science (Walter and Murphy, 2018). Its nature varies greatly depending on the context and underlying motivations, such as dread, wedge-driving, and wish-related motivations, adding further complexity to the development of universal fact-checking strategies. While much research has focused on general fact-checking practices, the comparative effectiveness of strategies tailored to specific underlying motivation types remains underexplored, highlighting an important area for future investigation.

Correction attributes of fact-checking posts

A correction in fact-checking is an attempt to change individuals' beliefs or understandings based on accurate information. Scholars have consistently examined the effectiveness of

correction attributes, focusing on content (Aruguete et al., 2023; Tandoc et al., 2022; Walter et al., 2020; Young et al., 2018) and source characteristics (Liu et al., 2023; Vraga and Bode, 2017; Zhang et al., 2021a). Some research has investigated the effects of emotional appeals (Lee et al., 2021; Lu and Shen, 2023) on user engagement with fact-checking information. However, little is understood about the favorability of these correction attributes, and whether people generally support these components. How these attributes influence endorsement through both "likes" and comments of fact-checking posts remains unclear. Our study explored two correction attributes that may convey credibility, including the source actor and the references used within the fact-checking message. Past research has established the role of credibility in persuasion (Kumkale et al., 2010; Pornpitakpan, 2004). We also investigate the potential impacts of different visual cues since the modalities of these cues may enhance the persuasiveness of the fact-checking messages through mechanisms such as increased attention and vividness.

Actor. In China, the actors involved in fact-checking include the government, media, and fact-checking organizations. According to previous research, perceived source credibility can be an explanatory mechanism for varied fact-checking effectiveness across different sources (Liu et al., 2023). For example, expertise is an important factor in measuring the credibility of a source. Experts, such as the Centers for Disease Control and Prevention (CDC), government health agencies, universities, and health institutions, are often more effective than non-experts due to their presumed superior health knowledge and skills (van der Meer and Jin, 2020; Vraga and Bode, 2017; Wang, 2023; Zhang et al., 2021a). Consequently, individuals are more likely to acknowledge and engage with such information, such as liking posts to express endorsement. For example, Wang (2023) found that people were more likely to click "like" on corrective messages from expert sources than on those from social peer sources.

Studies have compared the effectiveness of different types of sources in reducing individuals' intent to share misinformation, changing attitudes, and reducing misperceptions through experimental or survey methods (Liu et al., 2023; Vraga and Bode, 2017; Zhang et al., 2021a). However, in the context of social media corrections, researchers usually focus on the impact of sources on correction effectiveness, and few studies have explored how different actor types on Chinese social media affect endorsement through "likes" and expressed support in comments on fact-checking posts.

Reference. On social media, expert references are defined as citing individuals or organizations with expertise in a related field. These references are primarily categorized into two distinct types: specific and non-specific references (Chu et al., 2023; McGlynn et al., 2020). Specific references refer to the inclusion of verifiable information sources, such as People's Daily or the CDC, in a post to support the accuracy and credibility of the information. In contrast, non-specific references lack clear source attribution; ambiguous terms such as "experts" or "the relevant department" are used.

In the context of persuasion, the practice of associating with or referencing a source perceived as credible or authoritative is known as borrowed credibility (Chen et al., 2021). Even misinformation that incorporates fake experts, non-existing statistics, or fabricated evidence is perceived as more credible than misinformation without any

references (Hameleers, 2022). Compared to vague or non-specific references, posts with specific references, such as links to source material or citations of experts, are generally regarded as more credible and trustworthy. Specific and verifiable references enhance the likelihood of the content being accepted as reliable by readers, thereby increasing its potential for endorsement and sharing. In the overwhelming information landscape of social media, where users are inundated with content (Bontcheva et al., 2013), specific references play an even more critical role in guiding users to discern valuable information. Fact-checking posts with specific references can significantly improve users' truth discernment, thus enhancing users' perceived utility. This may lead users to "like" or make supportive comments on these posts as a form of social exchange, signifying their interest in and support for knowledge acquisition.

Visual cues. Visual cues refer to visual elements like images, videos, and linked news articles to guide attention, convey information, or prompt individuals to engage in specific cognitive activities. They are important factors influencing people's processing of information and its subsequent cognitive and behavioral outcomes (Powell et al., 2015; von Sikorski, 2022). On social media, gaining the attention and interest of users is typically the foundational premise for influencing people's cognition and behavior. Compared to plain text, content with visual cues is more likely to capture social media users' interest and attention (Li and Xie, 2020). The reasons could be that the human brain understands and processes visual information much faster than textual information, particularly for people who are new to the information (Mayer and Moreno, 2003), and that visual content, especially images and videos, can evoke emotional resonance, thereby increasing the impact of the information (Zhang et al., 2021b).

To effectively convey fact-checked content, the findings of a collaborative workshop with fact-checking professionals, communication experts, and researchers recommended using visual element graphs (Hettiachchi et al., 2023). In particular, social media managers and copy editors suggest the addition of visual storytelling in fact checks to engage with more audiences (Juneja and Mitra, 2022). In an empirical study, Young et al. (2018) found that video formats are more effective at capturing people's interest and attention, significantly reducing misperceptions compared with text-based fact-checking. Therefore, fact-checks with visual cues may be more likely to receive "likes" and supportive comments than text-only fact-checks. From both a theoretical and practical perspective, there is an increased need to understand the potential impacts of these visual cues.

Awareness prompts and legal warnings on social media

Prompts signaling awareness of misinformation highlights the importance of not blindly trusting unverified information and encouraging the public to avoid spreading potential misinformation (Pennycook and Rand, 2022; Pennycook et al., 2021). These messages often include slogans such as "the Internet is not a lawless place" and "do not create, believe, or spread rumors," advocating for a collective effort to create a safer cyberspace. Rather than targeting specific instances of misinformation, this strategy fosters public caution and enhances media and information literacy.

Awareness prompts offer advantages in countering misinformation. For example, they may increase the quality of information shared on social media by increasing individuals' awareness of the accuracy (Pennycook and Rand, 2022). Furthermore, these prompts may promote social cohesion by emphasizing shared societal values and collective responsibility (Zhu and Yang, 2023).

However, although intended to promote public vigilance, awareness prompts can sometimes be perceived as a directive imposed by authorities, triggering resistance or aversion (van der Meer and Hameleers, 2021). Their effectiveness may largely depend on tone and delivery; excessive force or perceived arrogance may provoke resistance. Thus, exploring their impact on endorsement levels and comments with fact-checking content is essential.

Legal warnings mention punishment for creating or spreading misinformation, as enforced by both law and social media platforms. "A guide to anti-misinformation actions around the world" highlights that China has some of the world's strictest laws against misinformation (Poynter, 2023), such as Article 291(1) of the Chinese Criminal Law (Criminal Law Amendment Act (9)), individuals who fabricate or deliberately spread misinformation online would face prison sentences (China Law Translate, 2015). Moreover, social media platforms have taken measures to address misinformation. The "Weibo Complaints Operating Rules" outline penalties for accounts spreading misinformation, such as the deduction of credit history points, temporary suspension from posting and being followed, and access restrictions (Weibo, 2021). Similar measures are evident globally, as Radu (2020) notes, many countries are fighting the infodemic by providing guidelines for social media companies to remove misinformation or by criminalizing malicious coronavirus falsehoods.

The use of legal warnings for addressing misinformation remains controversial. On one hand, the regulation of the information landscape is fragile and can easily lead to backfire (Caled and Silva, 2022). For example, these measures, intended to address misinformation, may infringe upon rights like freedom of expression and media freedom, as well as the potential legitimation of censorship systems (Zhu and Yang, 2023). Empirical studies, such as those by Tully et al. (2022), suggest skepticism toward legal interventions due to concerns over speech freedom restrictions and potential biases in law enforcement. Furthermore, legislative measures against misinformation are generally at a nascent stage, and it is currently too early to assess their negative and positive impacts (Haciyakupoglu et al., 2018).

On the other hand, research by Kozyreva et al. (2022) highlights a public preference for suppressing harmful misinformation over protecting free speech. Furthermore, the influence of the presumed influence of misinformation can enhance public support for governmental restrictions on social media misinformation (Luo and Cheng, 2021). In addition, when people observe punitive actions against illegal or unethical behavior involving the creation or dissemination of misinformation on social media, some may experience moral satisfaction, perceiving these actions as upholding societal moral and ethical standards (Kozyreva et al., 2023). This satisfaction is often expressed through social media endorsements reflecting support for measures that promote societal harmony and security. Thus, while legal warnings may evoke public support in certain contexts, their overall impact on endorsement and societal outcomes remains unclear.

Fact-checking across different misinformation types on social media

Prior research has categorized rumors into three distinct types: dread, wedge-driving, and wish rumors (Chua et al., 2017; Knapp, 1944; Song et al., 2021). Misinformation can be classified in a similar manner based on the underlying motivations behind why the misinformation was spread. Wish misinformation expresses hope for better outcomes, wedge-driving misinformation aims to create or deepen conflicts by targeting certain groups or individuals, and dread misinformation generates fear or anxiety (Chua et al., 2017; Knapp, 1944; Song et al., 2021).

Different motivations for misinformation trigger various cognitive and affective processes, as well as subsequent behaviors, which may influence individuals' acceptance and dissemination of information. For example, political misinformation aimed at provoking anger or aggression toward the opposing party (i.e. wedge-driving) may lead users to believe and spread these false claims, as anger is primarily associated with the behavioral tendency to attack (Nabi, 1999). To effectively counter misinformation, fact-checking information needs to be tailored based on the specific motivations behind different types of misinformation to better meet the needs of the audience.

However, there has been a lack of comprehensive studies on whether correction attributes, awareness prompts, and legal warnings have differential impacts on endorsement levels and the endorsement expressed in the comments across these three types of misinformation. Therefore, we propose the following research question:

> RQ1. How do correction attributes, awareness prompts, and legal warnings differentially impact endorsement through (a) "likes" and (b) support reflected in comments on fact-checking information across dread, wedge-driving, and wish misinformation?

Figure 1 illustrates our conceptual model depicting the relationship between counter misinformation strategies and two endorsement measures across dread, wedge-driving, and wish misinformation.

Method

Data collection

Weibo Piyao, established in 2010, is Weibo's first official account dedicated to fact-checking. It creates original fact-checking content and shares posts from other verified sources. Figure 2 depicts the process of data collection, cleaning, and processing. Our data collection encompassed all posts published by the Weibo Piyao account from 18 November 2010, to 31 May 2022. This dataset comprised 595 posts by "Weibo Piyao" and 12,034 reposts from other verified accounts. By tracing these reposts to their original sources, we collected 9,519 unique posts from the other verified accounts. After eliminating duplicated posts, we obtained 10,106 unique original posts.

In our manual review, we removed 970 posts that were not related to fact-checking, resulting in a total of 9,136 fact-checking posts. Then, we collected associated source information for these posts, including verification status, gender, number of fans,

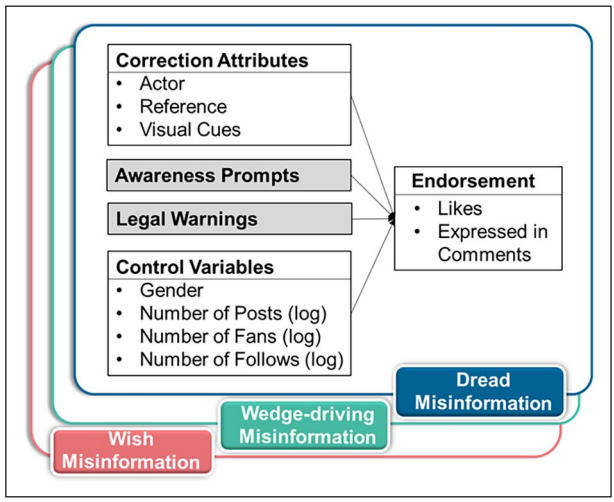

Figure 1. Conceptual Model Depicting the Relationship between Counter Misinformation Strategies and Two Endorsement Types across Dread, Wedge-Driving, and Wish Misinformation.

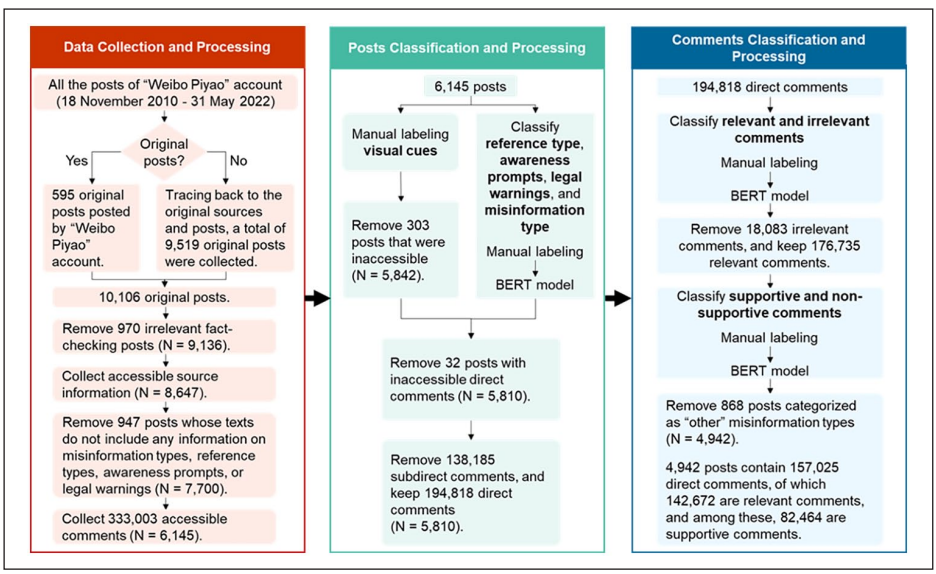

Figure 2. Process of Data Collection, Cleaning, and Processing.

number of follows, number of posts by the account, and a brief public profile, and obtained source information for 8,647 posts. A subset of 947 posts' texts did not include any information on misinformation types, reference types, awareness prompts,

or legal warnings (detailed examples can be found in Table S1), so we removed these posts and obtained 7,700 posts associated with the number of likes. Based on the sample of posts, we collected associated comments. Since some of these posts did not have any comments, we ultimately collected 6,145 posts with 333,003 comments.

During the manual labeling of visual cues, we removed 303 posts that were either inaccessible or had been deleted. In addition, we excluded 32 posts without direct comments and 868 posts categorized as "other" misinformation types. After preprocessing the dataset, our final dataset for regression analysis consisted of 4,942 posts and 157,025 direct comments.

To calculate the endorsement reflected in the comments, we manually labeled 2,000 comments and then used machine learning to identify whether the comments were relevant. For example, a comment would be removed if they used nonsensical language. This resulted in 176,735 relevant comments. Next, we manually labeled 1,500 comments and employed machine learning to identify whether the relevant comments expressed support for fact-checking information, ultimately identifying 103,090 supportive comments.

Measurement and coding scheme

The unit of analysis in this study is the individual post. Drawing on existing literature, we first developed initial coding protocols for each category. A pilot coding session was conducted using a sample of posts, during which two graduate students specializing in communication independently applied the coding protocol. Through iterative discussions, they resolved discrepancies and collaboratively refined the coding scheme. Once substantial agreement was achieved, as indicated by Cohen's kappa values exceeding 0.7, the two coders proceeded to independently label the remaining posts.

Misinformation type. We classified misinformation into four categories: (1) *dread misinformation*, which arouses feelings of worry or fear; (2) *wedge-driving misinformation*, which specifically targets certain groups or individuals, aiming to create or deepen tensions and conflicts by provoking strong emotional reactions against particular targets, such as aggressive urban management officials or police; (3) *wish misinformation*, which expresses hope and wishes, prompting audiences to hope for a better outcome; and (4) *other*, which does not fall into the first three categories (Cohen's kappa = 0.78). Examples of misinformation types are shown in Table S2.

Actor. We categorized account actors into six groups using Weibo's classification system, without applying manual labeling. (1) "*government*," which refers to government agencies, officials, or related departments; (2) "*media*," which includes news and media organizations or journalists; (3) "*fact-checking organizations*," which encompass non-governmental organizations, social groups, public welfare organizations, and academic institutions, predominantly specialized fact-checking organizations in our dataset; (4) "*companies*," which refers to businesses, corporations, and other commercial entities; (5) "*celebrities*," which includes individuals who built a reputation for their expertise in a specific field and are influential; and (6) "*schools*,"

which encompasses educational institutions such as universities, colleges, and schools at various levels. Examples of actor types are shown in Table S3.

Reference. According to previous studies (Chu et al., 2023; McGlynn et al., 2020), the reference in each post was labeled (1) *specific reference*, (2) *non-specific reference*, or (3) *no reference* based on the presence and specificity of the reference source (Cohen's kappa = 0.74). A specific reference indicates the inclusion of a verifiable source, such as a recognized institute (e.g. People's Daily, CDC) or individual. Conversely, a non-specific reference refers to vague citations of authority, such as "an expert," "relevant department," and "multiple sources." Examples of reference types are shown in Table S4.

Visual cues. Our dataset features six post formats: (1) *text only*, which contains only written content without any accompanying images or media; (2) *text + supplementary image*, which includes text along with an image that supports or enhances the content of the text; (3) *text + irrelevant image*, which includes text accompanied by an image that does not relate to the content of the text; (4) *text + text image*, where the text is supplemented with an image that contains additional text; (5) *text + video (link)*, which includes text and a link to a video; and (6) *text + news article (link)*, which features text along with a link to a news article (Cohen's kappa = 0.88). Examples of visual cues are shown in Table S5.

Awareness prompts. Posts prompting awareness of misinformation were defined as emphasizing the importance of not blindly trusting unverified information and encouraging the public to avoid spreading potential misinformation. Posts without a mention of misinformation awareness were labeled "*absent*"; otherwise, they were marked "*present*" (Cohen's kappa = 0.92). Examples of awareness prompts are shown in Table S6.

Legal warnings. Legal warnings refer to social media platform policies and legal consequences related to the creation or dissemination of misinformation in fact-checking posts. Posts lacking this content were labeled "*absent*," while those containing it were marked "*present*" (Cohen's kappa = 0.89). Examples of legal warnings are shown in Table S7.

Relevance of comments. Suppose a comment contains only any of the following elements. In that case, it is labeled an "*irrelevant comment*": mentioning other users without additional content, meaningless numbers or punctuation marks, expressions of support for a celebrity (e.g. fan support for celebrities), simple greetings, ambiguous or incomprehensible content, or requests for attention to other unrelated events. Otherwise, it would be marked as a "*relevant comment*" (Cohen's kappa = 0.80). Examples of relevance of the comments are shown in Table S8.

Endorsement expressed in comments. According to previous studies (Wang et al., 2022; Zubiaga et al., 2018), we categorized the endorsements expressed in the comments into two groups: (1) "*support*," which includes a clear inclination of support toward the fact-checking information or the actor, and (2) "*non-support*," which includes denying or

questioning fact-checking information or the actor, as well as ambiguous or neutral stances (Cohen's kappa = 0.78). Examples of endorsements reflected in the comments are shown in Table S9.

Supervised machine learning

We applied supervised machine learning to the six criteria that were manually coded: misinformation type, reference, awareness prompts, legal warnings, the relevance of the comments, and the endorsement expressed in comments. To facilitate this, we manually labeled datasets comprising 1,595 posts for the misinformation type classifier, 1,215 posts for the reference classifier, 1,332 posts for the awareness prompts and legal warnings classifiers, 2,000 comments for the relevance of the comment classifier, and 1,500 comments for the endorsement expressed in the comment classifier. We utilized bidirectional encoder representation from transformers (BERT), a large language model proposed by Devlin et al. (2018), for the classification process. To evaluate the performance of different classification methods, we randomly divided the samples of coded messages into training (80%) and testing sets (20%). Two common metrics are used to assess the performance of a machine learning classifier: precision and recall, which are often synthesized into a comprehensive statistic known as the F1 score. In this study, the precision, recall, and F1 scores for the classification task were all above 0.75, indicating that the machine learning classifier demonstrated good performance in accurately predicting the six dimensions in the analyzed social media posts, as shown in Table S10.

Analysis

This study examines how different strategies are associated with two types of endorsement of fact-checking posts. The actor, reference, visual cues, awareness prompts, and legal warnings were chosen as the independent variables, while meta-user information on Weibo (gender, number of posts, number of fans, and number of follows) was chosen as the control variable. The distribution of misinformation type and fact-checking strategies are shown in Figure 3, and the selection of regression models are justified in Table S11.

Results

The main results address RQ1a and RQ1b, illustrating how correction attributes, awareness prompts, and legal warnings influence endorsement – measured through "likes" and comments – across dread, wedge-driving, and wish misinformation. Endorsement through "likes" is presented in Figure 4/Table S12, while endorsement expressed in comments is shown in Figure 5/Table S13.

Endorsement through "likes"

To explore the impact of different actors on endorsement levels while fact-checking various types of misinformation, we observed that, compared with the government,

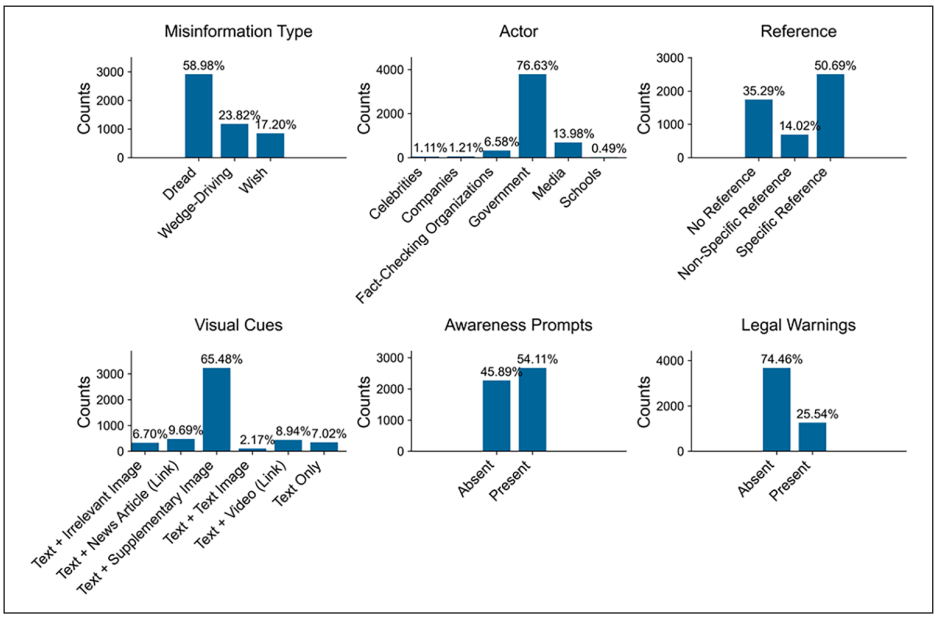

Figure 3. Distribution of Misinformation Type and Fact-Checking Strategies.

fact-checking information about wedge-driving misinformation posted by fact-checking organizations was found to be associated with significantly lower endorsement levels ($\beta = -0.56$, $p < .001$), while fact-checking information about wish misinformation posted by fact-checking organizations was associated with significantly higher endorsement levels ($\beta = 0.34$, $p = .045$). In contrast, fact-checking information published by media, companies, celebrities, and schools received significantly higher levels of endorsement than government information across all three types of misinformation.

When examining the impact of reference, we found that in the case of dread misinformation, fact-checking posts with specific references ($\beta = 0.44$, $p < .001$) or without references ($\beta = 0.32$, $p < .001$) received significantly greater endorsement than posts with a non-specific reference. However, in the case of wedge-driving misinformation, fact-checking posts with specific references ($\beta = -1.03$, $p < .001$) or without references ($\beta = -0.38$, $p < .001$) were found to be associated with significantly lower endorsement levels than posts with a non-specific reference.

For visual cues, compared with text combined with supplementary images, irrelevant images, video (link), and news article (link) were associated with significantly lower levels of endorsement across the three types of misinformation. Text combined with text images was also associated with lower endorsement in fact-checking information about dread ($\beta = -2.53$, $p < .001$) or wedge-driving misinformation ($\beta = -2.51$, $p < .001$), but it showed no significant difference from text only in the context of wish misinformation.

The use of awareness prompts in fact-checking information about dread misinformation significantly increased the level of endorsement ($\beta = 0.49$, $p < .001$), while the use of

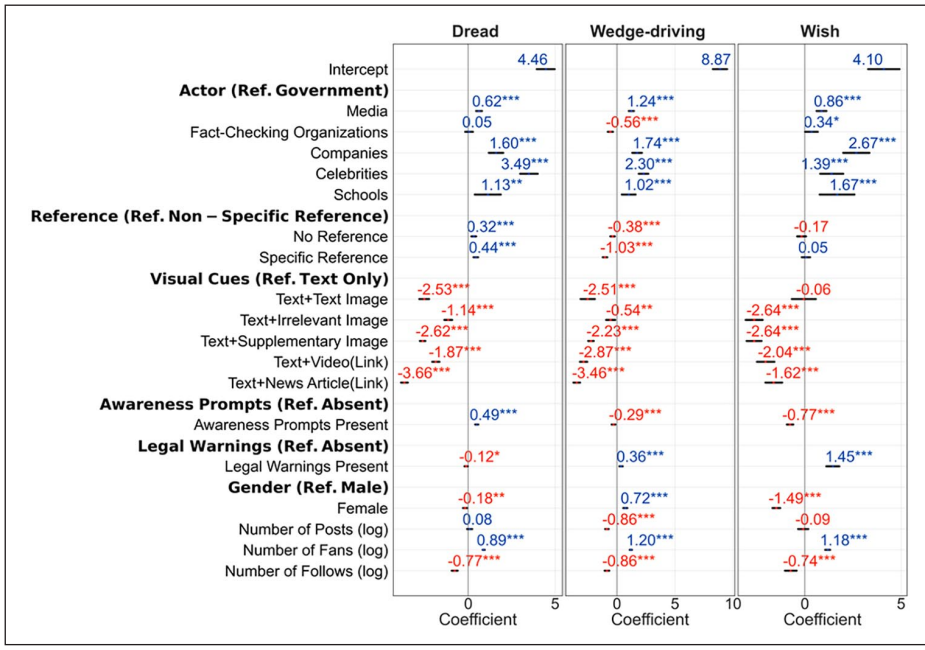

Figure 4. Negative Binomial Regression Results for Predicting "Liking" of Fact-Checking Information across Dread, Wedge-Driving, and Wish Misinformation.
Notes. Coefficients are color-coded, with red indicating a negative coefficient and blue indicating a positive coefficient. Error bars represent 95% confidence intervals.
*$p < .05$; **$p < .01$; ***$p < .001$.

awareness prompts in fact-checking information about wedge-driving misinformation ($\beta = -0.29$, $p < .001$) or wish misinformation ($\beta = -0.77$, $p < .001$) significantly decreased the level of endorsement.

When examining the impact of legal warnings, the use of legal warnings in fact-checking information about dread misinformation received significantly lower levels of endorsement ($\beta = -0.12$, $p = .014$), while the use of legal warnings in fact-checking information about wedge-driving misinformation ($\beta = 0.36$, $p < .001$) and wish misinformation ($\beta = 1.45$, $p < .001$) received significantly greater levels of endorsement.

Endorsement expressed in comments

Consistently, across the three types of misinformation, compared with text combined with supplementary images and text with a news article (link) received significantly more endorsements expressed in comments across dread, wedge-driving, and wish misinformation.

For other variables, there were inconsistent results across the three types of misinformation. For actors, compared with those of the government, media, fact-checking organizations, and companies received significantly less endorsement expressed in the

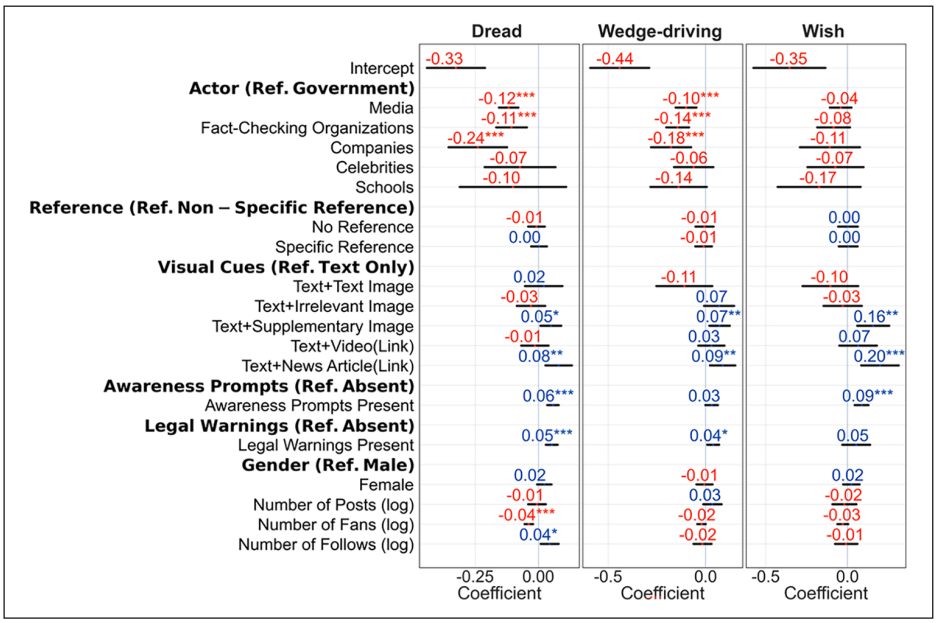

Figure 5. Linear Regression Results for Predicting Endorsement Expressed in the Comments of Fact-Checking Posts across Dread, Wedge-Driving, and Wish Misinformation.
Notes. Coefficients are color-coded, with red indicating a negative coefficient and blue indicating a positive coefficient. Error bars represent 95% confidence intervals.
*$p < .05$; **$p < .01$; ***$p < .001$.

comments across dread and wedge-driving misinformation. The presence of awareness prompts in fact-checking information received significantly more endorsement expressed in the comments across dread ($\beta = 0.06$, $p < .001$) and wish misinformation ($\beta = 0.09$, $p < .001$). When examining the impact of legal warnings, the presence of legal warnings in fact-checking posts about dread misinformation ($\beta = 0.05$, $p < .001$) and wedge-driving misinformation ($\beta = 0.04$, $p = .015$) received significantly greater endorsement in the comments.

Discussion and conclusion

This study investigated how correction attributes, awareness prompts, and legal warnings impact endorsement, measured through "likes" and comments, across three types of misinformation: wish, dread and wedge-driving. We found that both awareness prompts and legal warnings usually have a positive impact on endorsement expressed in comments. Posts with visual cues received fewer "likes," but adding visual cues such as supplementary images or news articles would significantly increase supportive comments. Fact-checking posts published by Chinese government accounts received fewer "likes." These key findings are shaped by the unique cultural, social, and political characteristics of the Chinese context.

The impact of awareness prompts and legal warnings on endorsement, as reflected in comments, is generally positive. This phenomenon can be explained by the alignment of these strategies with cultural and institutional values in China. On Weibo, awareness prompts, such as slogans like "Do not make, believe, or spread misinformation," resonate with the values long advocated by the government and mainstream media (Zhu and Yang, 2023). These slogans reinforce familiar norms, eliciting public support. Legal warnings may invoke a sense of social responsibility in Chinese users, prompting supportive comments such as, "Those who spread misinformation should be punished." This reflects an alignment with collective norms and a tendency to trust government-led initiatives.

Compared to text-only posts, any additional visual cue did not increase the number of "likes" across the three types of misinformation. This phenomenon could be attributed to cultural and platform-specific factors: first, textual information aligns better with ethos of fact-checking, as authoritative media in China, such as People's Daily and CCTV often delivers concise, text-based fact-checks. Audiences may therefore perceive textual information as more trustworthy and credible. Second, Weibo's interactive interface requires users to click on links or images to access additional content, which can divert attention from clicking the "like" button. For example, users who click on a link to read detailed fact-checking information may not return to the original post to "like" it.

However, we observed that fact-checking posts, including supplementary images or news articles, garnered more supportive comments than text-only posts across the three types of misinformation. Although this seems to contradict our previous conclusion, it can be reasonably explained. Supplementary images or news articles provide multiple dimensions of information, promoting deeper thinking among users and increasing their comprehensive understanding and willingness to comment. Previous studies have also shown that visual cues can enhance credibility (Hameleers et al., 2020) and mitigate audience misunderstandings (Young et al., 2018). Therefore, fact-checking posts containing visual cues may be more likely to receive supportive comments.

Fact-checking posts published by Chinese government received significantly fewer "likes," but greater endorsement in comments, especially when addressing dread-driven and wedge-driving misinformation. On Chinese social media, many government accounts are created by local authorities and focus on local issues, which may not attract a wider audience. In addition, many of their posts prioritize self-promotion over service delivery and tend to use formal, monotonous language (Zheng and Zheng, 2014). However, fact-checking posts published by the government received more supportive comments. Among the fact-checking accounts on Weibo, government accounts are mainly from law enforcement agencies such as the Public Security Bureau and the Internet Police. Users tend to be more cautious when commenting on these posts, avoiding critical or negative remarks for fear that such comments may be controlled or deleted. This cautious behavior reflects a combination of regulatory awareness and cultural norms. In a collectivist society like China, public trust in government tends to be higher, and citizens are generally more supportive of interventions that align with collective norms.

Fact-checks with references receive significantly different levels of "likes" across the three types of misinformation. Fact-checking posts with specific references receive

more "likes" only when countering dread misinformation. However, when countering wedge-driving misinformation, those with specific references received significantly fewer "likes" than did those with non-specific references. When dealing with dread misinformation, providing clear evidence and explanations can help users alleviate concerns and fears and increase their acceptance of fact-checking information. Wedge-driving misinformation targets certain groups or individuals and may inherently attribute responsibility to those targets (Song et al., 2021). Fact-checking efforts addressing such misinformation frequently involve specific references to these targets. However, using vague information sources in fact-checking wedge-driving misinformation can enhance neutrality and reduce perceived confrontation, potentially fostering a more positive reception. This approach helps individuals avoid being seen as adversarial, which can facilitate engagement and support.

It should be noted that some variables were negatively associated with the number of "likes" but positively associated with supportive comments. "Liking" can show endorsement, but individuals may not actually "like" the information shown. However, the added analysis of comments provides a more complete picture of how these features impact endorsement. For example, individuals may not prefer to "like" a post that threatens legal action on spreading dread misinformation but rather understand the importance of the issue and show their support through comments. Our study examines multiple dimensions of endorsement to provide more meaningful and valid results. "Likes" were more variable, and comments were less so, suggesting different motivations and responses in these forms of engagement. Future research should explore these reasons in depth to better understand the nuances of online engagement and endorsement.

When discussing the limitations of this study, it is important to acknowledge that our research is currently focused solely on the Weibo platform and guideline for tailoring fact-checking strategies to specific types of misinformation may not be applicable to other cultural contexts. However, the principles of tailoring fact-checking strategies to specific motivations underlying misinformation – such as dread, wedge-driving, and wish – can be applied across different sociocultural contexts. However, certain aspects, such as the use of legal warnings, are more prominent in the Chinese context due to unique regulatory and institutional norms. Further research in other cultural and political settings is necessary to validate and refine these conclusions, ensuring their broader applicability.

This study investigated how correction attributes (actor, reference type, visual cues), awareness prompts, and legal warnings impact endorsement, measured through "likes" and comments, across three types of misinformation: wish, dread, and wedge-driving. The findings highlight that both awareness prompts and legal warnings usually have a positive impact on endorsement reflected in comments. Posts with visual cues received fewer "likes," but adding visual cues such as supplementary images or news articles would significantly increase supportive comments. Fact-checking posts published by Chinese government accounts received fewer "likes."

This study highlights that misinformation correction strategies are not a one-size-fits-all approach. Tailoring fact-checking strategies to the underlying motivations of misinformation can enhance both credibility and public acceptance. This study provides practical insights for designing more effective fact-checking strategies to combat

misinformation. For dread-driven misinformation, incorporating specific references and awareness prompts can strengthen engagement and trust. For wedge-driving misinformation, legal warnings may help mitigate its impact, but excessive reliance on fact-checking organizations or awareness prompts should be avoided. For wish-driven misinformation, legal warnings can act as a deterrent; however, pairing text with irrelevant images should be minimized to maintain credibility. These findings offer actionable recommendations for fact-checkers, policymakers, and platform regulators to refine misinformation interventions and improve public trust in verified information.

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Supplemental material

Supplemental material for this article is available online.

References

Aruguete N, Bachmann I, Calvo E, et al. (2023) Truth be told: how "true" and "false" labels influence user engagement with fact-checks. *new media & society* 27: 1443–1464.

Bontcheva K, Gorrell G and Wessels B (2013) Social media and information overload: survey results. arXiv:1306.0813. Available at: http://arxiv.org/abs/1306.0813 (accessed 25 December 2023).

Borah P and Xiao X (2018) The importance of 'likes': the interplay of message framing, source, and social endorsement on credibility perceptions of health information on Facebook. *Journal of Health Communication* 23: 399–411.

Brenes Peralta CM, Sánchez RP and González IS (2022) Individual evaluation vs fact-checking in the recognition and willingness to share fake news about Covid-19 via Whatsapp. *Journalism Studies* 23(1): 1–24.

Caled D and Silva MJ (2022) Digital media and misinformation: an outlook on multidisciplinary strategies against manipulation. *Journal of Computational Social Science* 5(1): 123–159.

Chen A, Chen K, Zhang J, et al. (2023) When national identity meets conspiracies: the contagion of national identity language in public engagement and discourse about COVID-19 conspiracy theories. *Journal of Computer-Mediated Communication* 28(1): 034.

Chen S, Xiao L and Mao J (2021) Persuasion strategies of misinformation-containing posts in the social media. *Information Processing & Management* 58(5): 102665.

China Law Translate (2015) People's Republic of China criminal law amendment (9). *China Law Translate*. Available at: https://www.chinalawtranslate.com/criminal-law-amm-9/ (accessed 14 February 2024).

Chu J, Zhu Y and Ji J (2023) Characterizing the semantic features of climate change misinformation on Chinese social media. *Public Understanding of Science* 32(7): 845–859.

Chua AYK, Aricat R and Goh D (2017) Message content in the life of rumors: comparing three rumor types. In: *2017 Twelfth International Conference on Digital Information Management (ICDIM)*, Fukuoka, Japan, 12–14 September, pp. 263–268. New York: IEEE.

Courchesne L, Ilhardt J and Shapiro JN (2021) Review of social science research on the impact of countermeasures against influence operations. *Harvard Kennedy School Misinformation Review*. Epub ahead of print 13 September. DOI: 10.37016/mr-2020-79

Devlin J, Chang M-W, Lee K, et al. (2018) BERT: pre-training of deep bidirectional transformers for language understanding. arXiv:1810.04805. Available at: https://arxiv.org/abs/1810.04805

Freiling I, Krause NM, Scheufele DA, et al. (2021) Believing and sharing misinformation, fact-checks, and accurate information on social media: the role of anxiety during COVID-19. *new media & society* 25: 141–162.

Haciyakupoglu G, Hui JY, Suguna V, et al. (2018) *Countering Fake News: A Survey of Recent Global Initiatives*. Singapura: S. Rajaratnam School of International Studies— Nanyang Technological University.

Hameleers M (2022) Separating truth from lies: comparing the effects of news media literacy interventions and fact-checkers in response to political misinformation in the US and Netherlands. *Information, Communication & Society* 25(1): 110–126.

Hameleers M, Powell TE and Van Der Meer TGLA (2020) A picture paints a thousand lies? The effects and mechanisms of multimodal disinformation and rebuttals disseminated via social media. *Political Communication* 37(2): 281–301.

Hettiachchi D, Ji K, Kennedy J, et al. (2023) Designing and evaluating presentation strategies for fact-checked content. In: *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, 21 October 2023, pp. 751–761. Available at: http://arxiv.org/abs/2308.10220 (accessed 5 February 2024).

Juneja P and Mitra T (2022) Human and technological infrastructures of fact-checking. *Proceedings of the ACM on Human-computer Interaction* 6(CSCW2): 1–36.

Knapp RH (1944) A psychology of rumor. *Public Opinion Quarterly* 8(1): 22–37.

Kozyreva A, Herzog S, Lewandowsky S, et al. (2022) Free Speech Vs. Harmful Misinformation: Moral Dilemmas in Online Content Moderation. Available at: https://www.mpg.de/19855740/0208-bild-how-people-resolve-dilemmas-in-online-content-moderation-149835-x

Kozyreva A, Herzog SM, Lewandowsky S, et al. (2023) Resolving content moderation dilemmas between free speech and harmful misinformation. *Proceedings of the National Academy of Sciences* 120(7): 2210666120.

Kumkale GT, Albarracín D and Seignourel PJ (2010) The effects of source credibility in the presence or absence of prior attitudes: implications for the design of persuasive communication campaigns. *Journal of Applied Social Psychology* 40(6): 1325–1356.

Lee J, Kanthawala S, Britt BC, et al. (2021) Prevalence of anger, engaged in sadness: engagement in misinformation, correction, and emotional tweets during mass shootings. *Online Information Review* 46(3): 422–440.

Li Y and Xie Y (2020) Is a picture worth a thousand words? An empirical study of image content and social media engagement. *Journal of Marketing Research* 57(1): 1–19.

Liu X, Qi L, Wang L, et al. (2023) Checking the fact-checkers: the role of source type, perceived credibility, and individual differences in fact-checking effectiveness. *Communication Research*. Epub ahead of print 27 October. DOI: 10.1177/00936502231206419.

Liu Y and Zhou R (2022) "Let's check it seriously": localizing fact-checking practice in China. *International Journal of Communication* 16: 023.

Lu Y and Shen C (2023) Unpacking multimodal fact-checking: features and engagement of fact-checking videos on Chinese TikTok (Douyin). *Social Media + Society* 9(1): 20563051221150406.

Luo M, Hancock JT and Markowitz DM (2022) Credibility perceptions and detection accuracy of fake news headlines on social media: effects of truth-bias and endorsement cues. *Communication Research* 49(2): 171–195.

Luo Y and Cheng Y (2021) The presumed influence of COVID-19 misinformation on social media: survey research from two countries in the global health crisis. *International Journal of Environmental Research and Public Health* 18(11): 5505.

McGlynn J, Baryshevtsev M and Dayton Z (2020) Misinformation more likely to use non-specific authority references: Twitter analysis of two COVID-19 myths. *Harvard Kennedy School Misinformation Review* 1(3): 37.

Mayer RE and Moreno R (2003) Nine ways to reduce cognitive load in multimedia learning. *Educational Psychologist* 38(1): 43–52.

Messing S and Westwood S (2013) Selective exposure in the age of social media: endorsements Trump partisan source affiliation when selecting news online. *Communication Research* 41: 1042–1063.

Nabi RL (1999) A cognitive-functional model for the effects of discrete negative emotions on information processing, attitude change, and recall. *Communication Theory* 9(3): 292–320.

Nekmat E (2020) Nudge effect of fact-check alerts: source influence and media skepticism on sharing of news misinformation in social media. *Social Media + Society* 6(1): 205630511989732.

Pang H, Liu J and Lu J (2022) Tackling fake news in socially mediated public spheres: a comparison of Weibo and WeChat. *Technology in Society* 70: 102004.

Pennycook G and Rand DG (2022) Accuracy prompts are a replicable and generalizable approach for reducing the spread of misinformation. *Nature Communications* 13(1): 2333.

Pennycook G, Epstein Z, Mosleh M, et al. (2021) Shifting attention to accuracy can reduce misinformation online. *Nature* 592(7855): 590–595.

Pornpitakpan C (2004) The persuasiveness of source credibility: a critical review of five decades' evidence. *Journal of Applied Social Psychology* 34(2): 243–281.

Powell TE, Boomgaarden HG, De Swert K, et al. (2015) A clearer picture: the contribution of visuals and text to framing effects. *Journal of Communication* 65(6): 997–1017.

Poynter (2023) A guide to anti-misinformation actions around the world. *Poynter*. Available at: https://www.poynter.org/ifcn/anti-misinformation-actions/ (accessed 21 November 2023).

Radu R (2020) Fighting the 'infodemic': legal responses to COVID-19 disinformation. *Social Media + Society* 6(3): 2056305120948190.

Rodrigues UM and Xu J (2020) Regulation of COVID-19 fake news infodemic in China and India. *Media International Australia* 177(1): 125–131.

Song Y, Kwon KH, Lu Y, et al. (2021) The "parallel pandemic" in the context of China: the spread of rumors and rumor-corrections during COVID-19 in Chinese social media. *American Behavioral Scientist* 65(14): 2014–2036.

Sumner EM, Ruge-Jones L and Alcorn D (2018) A functional approach to the Facebook like button: an exploration of meaning, interpersonal functionality, and potential alternative response buttons. *new media & society* 20(4): 1451–1469.

Tandoc EC, Lee JCB, Lee S, et al. (2022) Does length matter? The impact of fact-check length in reducing COVID-19 vaccine misinformation. *Mass Communication and Society* 27: 679–709.

Tully M, Madrid-Morales D, Wasserman H, et al. (2022) Who is responsible for stopping the spread of misinformation? Examining audience perceptions of responsibilities and responses in six sub-Saharan African countries. *Digital Journalism* 10(5): 679–697.

van der Meer TGLA and Hameleers M (2021) Fighting biased news diets: using news media literacy interventions to stimulate online cross-cutting media exposure patterns. *new media & society* 23(11): 3156–3178.

van der Meer TGLA and Jin Y (2020) Seeking formula for misinformation treatment in public health crises: the effects of corrective information type and source. *Health Communication* 35(5): 560–575.

von Sikorski C (2022) Visual polarisation: examining the interplay of visual cues and media trust on the evaluation of political candidates. *Journalism* 23(9): 1900–1918.

Vraga EK and Bode L (2017) Using expert sources to correct health misinformation in social media. *Science Communication* 39(5): 621–645.

Vraga EK and Bode L (2020) Defining misinformation and understanding its bounded nature: using expertise and evidence for describing misinformation. *Political Communication* 37(1): 136–144.

Walter N and Murphy ST (2018) How to unring the bell: a meta-analytic approach to correction of misinformation. *Communication Monographs* 85(3): 423–441.

Walter N, Cohen J, Holbert RL, et al. (2020) Fact-checking: a meta-analysis of what works and for whom. *Political Communication* 37(3): 350–375.

Wang X, Chao F, Yu G, et al. (2022) Factors influencing fake news rebuttal acceptance during the COVID-19 pandemic and the moderating effect of cognitive ability. *Computers in Human Behavior* 130: 107174.

Wang Y (2023) Engaging with misinformation and misinformation corrective messages on social media: examining the role of source cues, social endorsement cues, and prior attitudes. *Atlantic Journal of Communication* 32: 499–511.

Weibo (2021) Weibo complaints operating rules. Available at: https://kefu.weibo.com/faqdetail?id=20543 (accessed 12 July 2024).

Young DG, Jamieson KH, Poulsen S, et al. (2018) Fact-checking effectiveness as a function of format and tone: evaluating FactCheck.org and FlackCheck.org. *Journalism & Mass Communication Quarterly* 95(1): 49–75.

Zhang J, Featherstone JD, Calabrese C, et al. (2021a) Effects of fact-checking social media vaccine misinformation on attitudes toward vaccines. *Preventive Medicine* 145: 106408.

Zhang X, Tan H, Yin C, et al. (2021b) Does an image facilitate the sharing of negative news on social media? An experimental investigation. *Library & Information Science Research* 43(4): 101120.

Zheng L and Zheng T (2014) Innovation through social media in the public sector: information and interactions. *Government Information Quarterly* 31: S106–S117.

Zhu X and Yang S (2023) *Toward a Sociotechnical Framework for Misinformation Policy Analysis.in the Usage and Impact of Icts during the COVID-19 Pandemic*. 1st ed. Routledge, pp. 11–45. Available at: https://www.taylorfrancis.com/books/9781003231769 (accessed 26 December 2023).

Zubiaga A, Kochkina E, Liakata M, et al. (2018) Discourse-aware rumour stance classification in social media using sequential classifiers. *Information Processing & Management* 54(2): 273–290.

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